

FINAL LAB EXAM

(SE-213) Data Structure Algorithm & Application

OBJECTIVE

The objective of the Final Lab is to evaluate students on the overall knowledge and skills gained throughout all lab sessions of the course. This session consolidates learning by requiring students to demonstrate mastery of **Data Structure Algorithm and Applications** through integrated tasks that reflect real-world problem-solving.

EVALUATION METHOD

Students will be assessed using the **Final Lab Rubrics (5 marks)**, consisting of **five criteria**, each marked on the basis of **0 or 1**. Marks will be awarded based on task completion, correctness, efficiency, and clarity of the solution.

EVALUATION DETAILS

A Software House records **employee IDs** for attendance:

- Insert employee IDs into a **Doubly Linked List** at beginning.
- Delete IDs of employees who have left.
- Convert remaining IDs into a **Circular Linked List** to keep rotating attendance for daily verification.
- Sort employee IDs using **Merge Sort**.
- Show the list and sorted IDs after each step.

NED University of Engineering & Technology
Department of Software Engineering
FINAL EXAMS RUBRICS
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Criteria	Achieved (1)	Not Achieved (0)	Marks
Problem Identification	Able to correctly identify the problem(s) and all required data (input and output)	Not able to correctly identify the problem(s) and all required data (input and output).	
Problem Solving Skills	Uses an effective strategy to solve the problem(s) and clearly identifies the steps	Uses an ineffective strategy to solve the problem(s) and completely misses the steps	
Implementation & Debugging	Efficient implementation and excellent debugging skills. Solution works without any error	Incomplete implementation and ineffective debugging skills. Major errors in the implementation.	
Deliverable(s)	Generates correct and complete output.	Cannot generate the desired output.	
Depth of Knowledge	Thorough grasp of the concepts covered in the lab and can apply it in real-life situations	Does not grasp the concepts covered in the lab and cannot identify any real-life application	
Weighted CLO Score			
Remarks and Signature			